LANDVIEW BUYBACK SYSTEM

MVP Technical Specification Document

Internal Investment Buyback Management System

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Status: Ready for Implementation

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1. Executive Summary

The Landview Buyback System is an internal tool designed to manage land investment buyback agreements for Landview company. This system enables accountants and administrators to track client investments, calculate returns based on interest rates, manage maturity dates, and automate payment notifications.

1.1 Business Objectives

- Track and manage client land investment buyback agreements
- Automate ROI calculations based on investment duration and interest rates
- Send automated email notifications for maturity dates and payment reminders
- Provide comprehensive audit trails for all system actions
- Enable AI-powered document processing for client onboarding
- Streamline payment processing workflow for accountants

2. System Overview

The Landview Buyback System is a web-based platform that manages the entire lifecycle of land investment buyback agreements from client registration to final payment completion.

2.1 Key System Components

- User Authentication & Authorization Module
- Investment Management Module
- ROI Calculation Engine
- Email Notification System
- AI Document Processing Module
- Audit Logging System
- Reporting & Analytics Dashboard

3. User Roles & Permissions

3.1 Role Hierarchy

Super Admin

Permissions:

- Full system access and configuration
- Create, modify, and delete Admin and Accountant users
- View all audit logs and system activities
- Access all investment records
- Receive daily email summaries of pending payments
- Configure system settings and interest rates
- Export and analyze system-wide reports

Admin

Permissions:

- Create and manage Accountant users
- View and edit investment records
- Process investment extensions
- Receive daily and weekly email notifications
- Access audit logs for their actions
- Generate reports

Accountant

Permissions:

- Create new investment records
- View assigned investment records
- Mark payments as completed
- Process investment extensions
- Upload client documents via AI processing
- View payment due lists

4. Core Features & Functionality

4.1 Investment Management

4.1.1 Create Investment Record

Allows authorized users to create new investment buyback records with the following details:

- Transaction Date
- Client Name
- Plot Number/Size
- Duration (e.g., 6 months, 12 months)
- Maturity Date (auto-calculated)
- Principal Amount
- Interest Rate Percentage
- ROI Amount (auto-calculated)
- Upfront Payment Amount (if applicable)
- Maturity Amount (auto-calculated)
- Client Email
- Realtor Name
- Realtor Email

4.1.2 ROI Calculation Engine

The system automatically calculates:

- ROI = Principal × Interest Rate Percentage
- Maturity Amount = Principal + ROI - Upfront Payment
- Maturity Date = Transaction Date + Duration
- Days Until Maturity (real-time countdown)

4.1.3 Investment Extension

Users can extend an investment with the following functionality:

- Click 'Extend' button on investment record
- Enter new duration period
- System recalculates new maturity date
- Option to adjust interest rate for extension period
- Extension is logged in audit trail
- Client receives email notification of extension

4.1.4 Payment Completion

When an investment matures and payment is made:

- Accountant marks investment as 'Payment Initiated'
- Admin/Super Admin verifies and marks as 'Payment Completed'
- System logs both user actions with timestamps
- Status changes from 'Active' to 'Completed'
- Client receives payment confirmation email

5. Data Model & Database Schema

5.1 Core Tables

5.1.1 Users Table

Field Name	Data Type	Description
user_id	UUID, Primary Key	Unique user identifier
email	String, Unique, Required	User login email
password_hash	String, Required	Hashed password
full_name	String, Required	User's full name
role	Enum, Required	super_admin, admin, accountant
is_active	Boolean, Default: true	Account active status
created_at	Timestamp	Account creation date
created_by	UUID, Foreign Key	User who created account

5.1.2 Investments Table

Field Name	Data Type	Required/Optional
investment_id	UUID, Primary Key	Required
transaction_date	Date	Required
client_name	String	Required
plot_number	String/Number	Required
duration	String	Required
maturity_date	Date (Calculated)	Auto-calculated
principal	Decimal(15,2)	Required
interest_rate	Decimal(5,2)	Required
roi_amount	Decimal(15,2)	Auto-calculated
upfront_payment	Decimal(15,2)	Optional
maturity_amount	Decimal(15,2)	Auto-calculated
client_email	String	Optional

realtor_name	String	Required
realtor_email	String	Required
status	Enum	active, completed, extended
payment_initiated_by	UUID, Foreign Key	Optional
payment_completed_by	UUID, Foreign Key	Optional
created_at	Timestamp	Auto-generated
created_by	UUID, Foreign Key	Required

5.1.3 Audit Logs Table

Field Name	Data Type	Description
log_id	UUID, Primary Key	Unique log identifier
user_id	UUID, Foreign Key	User who performed action
action_type	String	CREATE, UPDATE, DELETE, etc.
entity_type	String	investment, user, etc.
entity_id	UUID	ID of affected entity
old_values	JSON	Previous values before change
new_values	JSON	New values after change
ip_address	String	User's IP address
created_at	Timestamp	When action occurred

6. Email Automation System

6.1 Email Types

6.1.1 Maturity Notification (To Client)

- Trigger: Investment reaches maturity date
- Recipient: Client email
- Content: Investment details, maturity amount, next steps
- CC: Realtor email

6.1.2 Daily Payment Due List (To Super Admins)

- Trigger: Daily at 8:00 AM
- Recipients: All Super Admin users
- Content: List of investments maturing today with client details

6.1.3 Weekly Reminder (To Admins)

- Trigger: Every Monday at 8:00 AM
- Recipients: Admin and Super Admin users
- Content: Investments maturing this week, pending payments

6.1.4 Extension Confirmation (To Client)

- Trigger: Investment extension processed
- Recipient: Client email
- Content: New maturity date, updated terms

6.1.5 Payment Completion (To Client)

- Trigger: Payment marked as completed
- Recipient: Client email
- Content: Payment confirmation, amount paid, thank you message

6.2 Email Service Provider

Recommended: SendGrid, AWS SES, or Mailgun with template management

7. AI Document Upload Feature

7.1 Purpose

Allow accountants to upload client-filled forms/documents that the system will automatically parse to create new investment records.

7.2 Workflow

13. User clicks 'AI Upload' button
14. Uploads PDF/Image of client buyback form
15. System uses OCR + AI (Claude API or similar) to extract data
16. System presents extracted data in editable form
17. User reviews, corrects any errors
18. User submits to create investment record

7.3 Technical Implementation

- Use Anthropic Claude API for document extraction
- Accept PDF, PNG, JPG file formats
- Extract: Client name, plot number, principal, interest rate, duration, emails
- Show confidence scores for extracted fields
- Store original document with investment record

8. Audit Logging System

8.1 Actions to Log

- User login/logout
- Create investment record
- Update investment record
- Delete investment record
- Extend investment
- Mark payment as initiated
- Mark payment as completed
- Create/edit/delete user
- Failed login attempts
- Email sent

8.2 Log Information Captured

- Timestamp
- User who performed action
- Action type
- Entity affected (investment ID, user ID, etc.)
- Old values (before change)
- New values (after change)
- IP address
- User agent (browser/device info)

9. Technical Architecture

9.1 Recommended Technology Stack

Backend

- Node.js with Express.js OR Python with FastAPI
- PostgreSQL database
- Prisma ORM (for Node.js) or SQLAlchemy (for Python)
- JWT for authentication

Frontend

- React.js with TypeScript
- Tailwind CSS for styling
- React Router for navigation
- Axios for API calls
- Recharts or Chart.js for data visualization

Email Service

- SendGrid, AWS SES, or Mailgun
- Email templates using Handlebars or similar

AI/ML

- Anthropic Claude API for document parsing
- PDF parsing libraries (pdf-parse, PyPDF2)

Infrastructure

- Docker for containerization
- AWS, DigitalOcean, or Vercel for hosting
- Redis for caching and job queues
- Bull or node-cron for scheduled jobs

10. User Interface Requirements

10.1 Dashboard

- Total active investments
- Total value of active investments
- Investments maturing this week
- Overdue payments
- Recent activity log
- Quick action buttons (New Investment, AI Upload)

10.2 Investment List View

- Filterable table (by status, date range, client name)
- Sortable columns
- Search functionality
- Pagination
- Bulk export to Excel/CSV
- Status indicators (Active, Matured, Completed, Extended)
- Action buttons (View, Edit, Extend, Mark Paid)

10.3 Investment Detail View

- All investment details in organized sections
- Timeline of actions (created, extended, paid)
- Days until maturity countdown
- Related documents (uploaded forms)
- Email history for this investment
- Audit log for this investment

10.4 Audit Log View (Super Admin Only)

- Filterable by user, action type, date range
- Detailed view showing before/after values
- Export functionality
- Visual timeline of system activities

11. Security Requirements

11.1 Authentication & Authorization

- Strong password requirements (minimum 12 characters, mixed case, numbers, symbols)
- JWT tokens with 24-hour expiration
- Refresh tokens for session management
- Role-based access control (RBAC)
- Account lockout after 5 failed login attempts

11.2 Data Protection

- All passwords hashed with bcrypt (minimum 12 salt rounds)
- HTTPS/SSL for all communications
- Database encryption at rest
- Input validation and sanitization on all forms
- SQL injection prevention (use parameterized queries)
- XSS protection
- CSRF protection

11.3 File Upload Security

- File type validation (only PDF, PNG, JPG allowed)
- File size limits (max 10MB)
- Virus scanning on uploaded files
- Files stored in secure cloud storage (AWS S3 with access controls)

11.4 Backup & Recovery

- Daily automated database backups
- 30-day backup retention
- Tested disaster recovery plan

12. Implementation Roadmap

Phase 1: Core Infrastructure (Week 1-2)

- Set up development environment
- Database schema design and creation
- User authentication system
- Role-based access control
- Basic frontend structure

Phase 2: Investment Management (Week 3-4)

- Create investment record functionality
- ROI calculation engine
- Investment list and detail views
- Edit and update functionality
- Investment extension feature
- Dashboard with basic metrics

Phase 3: Audit Logging (Week 5)

- Implement audit logging for all actions
- Audit log viewer for Super Admin
- Filtering and search functionality

Phase 4: Email Automation (Week 6)

- Email service integration
- Email templates creation
- Maturity notification system
- Daily payment due notifications
- Weekly reminder system
- Scheduled job setup (cron jobs)

Phase 5: AI Document Upload (Week 7-8)

- File upload interface
- Claude API integration
- Document parsing and data extraction
- Review and correction interface
- Document storage system

Phase 6: Testing & Refinement (Week 9-10)

- Comprehensive testing (unit, integration, E2E)
- User acceptance testing
- Performance optimization
- Security audit
- Bug fixes and refinements

Phase 7: Deployment & Training (Week 11-12)

- Production environment setup
- Data migration (if applicable)
- User training sessions
- Documentation creation
- Production deployment
- Monitoring setup

13. API Endpoints Specification

13.1 Authentication Endpoints

POST /api/auth/login
Request: { email, password }
Response: { token, user { id, email, role, full_name } }

POST /api/auth/logout
Invalidates the current JWT token

13.2 Investment Endpoints

GET /api/investments
Query params: status, page, limit, search, sort_by, order
Response: { investments: [], total, page, limit }

GET /api/investments/:id
Response: { investment details with all fields }

POST /api/investments
Request: All investment fields
Response: { created investment with auto-calculated fields }

PUT /api/investments/:id
Request: Fields to update
Response: { updated investment }

POST /api/investments/:id/extend
Request: { new_duration, new_interest_rate (optional) }
Response: { updated investment with new maturity date }

POST /api/investments/:id/mark-payment-initiated
Response: { updated investment }

POST /api/investments/:id/mark-payment-completed
Response: { updated investment, triggers completion email }

13.3 AI Upload Endpoint

POST /api/investments/ai-upload
Request: multipart/form-data with document file
Response: { extracted_data: { all investment fields }, confidence_scores }

13.4 Audit Log Endpoints

GET /api/audit-logs
Query params: user_id, action_type, entity_type, start_date, end_date, page, limit
Response: { logs: [], total, page, limit }

13.5 User Management Endpoints

GET /api/users

Response: { users: [] } (Super Admin and Admin only)

POST /api/users
Request: { email, password, full_name, role }
Response: { created user }

14. Conclusion & Next Steps

This MVP document provides a comprehensive blueprint for implementing the Landview Buyback System. The system will streamline investment tracking, automate notifications, and provide robust audit trails for accountability.

Immediate Action Items

19. Review and approve this specification document
20. Set up development environment and infrastructure
21. Begin Phase 1 implementation
22. Schedule weekly progress review meetings
23. Assign development team members to specific modules

Success Metrics

- 100% of maturing investments trigger automated emails
- All system actions logged with 100% accuracy
- AI document upload achieves 90%+ accuracy on field extraction
- Zero unauthorized access incidents
- System uptime of 99.5%+

Document prepared for Claude Code implementation

Ready for immediate development